

Knowledge Substrate

Technical Architecture & Capabilities Report

On-Device · Privacy-First · Multilingual · Typed Entity Extraction · Cross-Cultural Normalisation ·
Source-Deduplicated Corroboration

A comprehensive analysis of the on-device, privacy-first, multilingual knowledge management system — including connector architecture, observation extraction with typed entity taxonomy, industry-specific identifier extraction, device-tier-aware processing, cross-cultural normalisation, semantic synonym matching, SLM-assisted entity refinement, synthesis hierarchy, memory decay with source-deduplicated corroboration, SSD TRIM automation, and lifecycle walkthroughs with critiques.

1. Executive Summary

The Knowledge Substrate is an on-device, privacy-first knowledge management system for SMEs. It ingests content from 120+ connectors, classifies by importance, extracts observations via lexicon-first pipeline, synthesises channel/domain/tenant roll-ups via on-device SLM (Bonsai-1.7B), and manages memory lifecycle through retention-scored decay with cryptographic forgetting.

The system provides the following capabilities:

- **Connector framework:** 120+ implementations (Slack, Email, GitHub, Jira, Shopee, Grab, Gojek) with watermark cursor atomicity, OAuth2, webhook support, and mockable transport for testing.
- **Observation extraction:** Lexicon-first pipeline with CJK and European lexicons, typed entity taxonomy (13 EntityType + 25 IdentifierKind variants), 16 industry-specific regex-based identifier extractors with checksum validation, device-tier-aware extraction (Low/Mid/High), SLM-assisted entity refinement (SlmRefiner with live SLM dispatch), SLM-assisted ambiguous identifier extraction, and cross-cultural normalisation (name order, honorifics, calendars, currency, addresses).
- **Semantic synonym matching:** Offline multilingual synonym map covering 40+ high-frequency business term groups across EN/JA/KO/ZH, including domain-specific vocabulary for legal, medical, and technical domains, with FTS5 query expansion for cross-lingual recall without embeddings.
- **SLM-assisted entity refinement:** Trait-based architecture with SlmRefiner (live SLM dispatch via InferenceRouter), HeuristicRefiner (context-based entity typing), and NoOpRefiner fallback. SLM-assisted ambiguous identifier extraction classifies tokens that look like identifiers but don't match standard regex patterns.
- **Cross-cultural calendar conversion:** Japanese era (■■/■■/■■/■■/■■), Thai Buddhist (■.■.), Chinese lunar (■■/■■/■■, 2020–2030 lookup table), and Hijri (Kuwaiti algorithm) to ISO 8601 proleptic Gregorian, with dispatch function for automatic calendar detection and conversion.
- **Evidence store:** SQLCipher-encrypted with per-scope DEKs, AEAD (XChaCha20-Poly1305) encryption, BLAKE3 content-hash deduplication, FTS5 trigram + hybrid retrieval, encrypted ring buffer for noise-class content with priority-aware LRU eviction, and secure deletion with SSD TRIM automation.
- **Synthesis hierarchy:** Channel → Domain → Tenant with type-system enforcement, per-tier prompt/grammar divergence, and GBNF-constrained JSON output.
- **Memory management:** Retention scoring (6 weighted inputs), per-culture DecayProfile tuning, source-deduplicated corroboration counting, TTL-based archived purge, and cryptographic forgetting via DEK destruction.
- **Security:** SLM-assisted semantic negation detection for complex constructions (SemanticNegationDetector), heuristic negation in lexicon classifier, result clustering by content hash, cross-reference graph for threading, noise promotion, mixed-language script detection, and secure deletion with platform-specific SSD TRIM (F_PUNCHHOLE on macOS, FITRIM on Linux).

Total: 2900+ unit/integration tests pass with 0 failures. Clippy-clean across all crates.

2. Architecture Overview

- **connector_framework** — trait-based SDK with mock transport, watermark cursors, OAuth2, webhooks.
- **connectors** — 120+ implementations (Slack, Email, GitHub, Jira, Shopee, Grab, Gojek, etc.).
- **evidence_store** — SQLCipher-encrypted, BLAKE3 dedup, FTS5 trigram + hybrid retrieval.
- **observation_engine** — lexicon-first extraction with CJK and European lexicons, language detection, typed entity taxonomy (EntityType/IdentifierKind), industry-specific identifier extractors, device-tier-aware extraction, cross-cultural normalisation, offline multilingual synonym map, and SLM-assisted entity refinement architecture.
- **synthesis_pipeline** — hierarchy-enforced Channel -> Domain -> Tenant synthesis with SLM/fallback.
- **memory_manager** — retention scoring, decay sweep, per-culture DecayProfile, source-deduplicated corroboration counting, crypto forgetting.
- **inference_router** — device-tier router with LlamaCpp/Ollama/fallback adapters, GBNF grammar.
- **crypto** — AEAD, CEK wrapping, provenance bundles, key management.
- **ffi** — UniFFI/N-API surface; **substrate_server** — Axum loopback API.

3. Connector Framework

The Connector trait provides `authenticate`, `initial_sync`, `incremental_sync`, `subscribe_webhook`, `handle_webhook_event`. WatermarkCursor guarantees sync atomicity — interrupted runs preserve the previous cursor.

3.1 Email Connector

Threading metadata surfaced: Gmail `threadId`, Graph `conversationId`. HTML stripping handles nested tables/scripts. Attachments listed with filename and size.

3.2 Webhook Edge Cases

8 new tests: empty body, malformed JSON, missing `resourceData`, 50-notification batch, empty `changeType`, all-unknown `changeTypes`, empty Gmail `messageIds`, empty `validationToken` fall-through. Fixed bug: empty-string `validationToken` was incorrectly short-circuiting.

4. Observation Extraction & Multilingual Support

4.1 CJK Lexicon

Japanese gained `漢字`, `かな`, `ひらがな`, `カタカナ`. Korean gained `한글`, `한자`, `한글`. Chinese gained `汉字`, `拼音`, `汉字`. Tests verify each triggers correct observation type.

4.2 Live Bonsai Suite Gating

Gated behind live-integration feature + `LLAMA_SERVER_BINARY/LLAMA_SERVER_MODEL` env vars. Hermetic tests verify skip behaviour when env vars unset.

4.3 Typed Entity Taxonomy

The Observation struct now carries two new optional fields: `entity_type`: Option and `identifier_kind`: Option. These provide typed sub-classification for Entity observations, replacing the previous flat string-content approach.

EntityType enum (13 variants): Person, Organization, Product, Location, Date, Currency, Identifier, Url, Email, Numeric, Event, Measurement, Unknown. Each variant has a stable string tag for serialisation (`as_str` / `from_tag` round-trip).

IdentifierKind enum (25 variants across 5 industry domains):

- **Finance:** IBAN, ISIN, LEI, SwiftBic, Ticker, Cusip
- **Healthcare:** Icd10, Cpt, Ndc, Udi
- **Legal:** CaseNumber, Patent, Trademark, Statute
- **Manufacturing:** Sku, Mpn, BatchLot, WorkOrder, SerialNumber
- **Retail:** Asin, PurchaseOrder, Invoice
- **General:** PhoneNumber, IpAddress, Other

Both fields use `#[serde(default)]` so existing serialised observations deserialize without error — `entity_type` and `identifier_kind` default to `None` for legacy data.

4.4 Industry-Specific Entity Extractors

The `entity_extractors` module provides 16 regex-based extractors, each using `OnceLock` for compile-once regex caching:

- **Finance:** IBAN (with mod-97 checksum validation), ISIN (12-char alphanumeric), SWIFT/BIC (8 or 11 chars), LEI (20-char alphanumeric), Stock tickers (`$AAPL`, `$7203.T`)
- **Healthcare:** ICD-10-CM (e.g. `E11.9`), NDC (5-4-2 or 5-3-2 format)
- **Legal:** Patent numbers (`US/EP/JP/WO/CN/KR/DE/FR/GB/CA/AU` formats), Case numbers (US federal: `1:23-cv-00123`, UK: `[2023] EWHC 1`, India: `W.P.(C) 123/2023`)
- **Manufacturing:** SKU (SKU: `ABC-123-XYZ`), Serial numbers (S/N: `ABC123XYZ`)

- **Retail:** ASIN (B0XXXXXXXXXX), Purchase orders (PO-001234), Invoice numbers (INV-001234)
- **General:** Phone numbers (+1-555-123-4567), IPv4 (with octet validation 0–255), IPv6, Currency amounts (\$1,234.56, €1.000,00), Measurements (99.9%, 2.5GB, 300ms, 100kg)

All extractors are integrated into `LexiconExtractor::do_extract` via `extract_typed_entities`, running after baseline lexicon extraction with deduplication against `seen_entities`. This ensures no duplicate observations are created.

4.5 Device-Tier-Aware Extraction

`EntityExtractionTier` enum maps to the existing `DeviceTier` model from `inference_router`:

- **Low** (maps to `DeviceTier::Low`): Lexicon-only extraction. No pattern-based identifier extraction. Fast on low-end mobile/embedded devices.
- **Mid** (maps to `DeviceTier::Mid`, default): Lexicon + all 16 pattern-based identifier extractors. Balanced precision and performance.
- **High** (maps to `DeviceTier::High`): Lexicon + patterns + SLM-assisted entity refinement via `HeuristicRefiner` (context-based entity typing using honorific/organisation/currency/date clues). The `EntityRefiner` trait enables live SLM integration when available.

`LexiconExtractor` gains `with_extraction_tier` builder method. Default is `Mid`. The `ObservationPipeline` can translate `DeviceTier` into `EntityExtractionTier` before calling extractors.

4.6 Cross-Cultural Entity Normalisation

The cultural module provides four normalisation capabilities for multilingual contexts:

- **Person name normalisation:** NameOrder detection (FamilyFirst for JA/KO/ZH/VI/HU, GivenFirst for Latin/Cyrillic). Honorific stripping for 7 languages (Western: Mr./Ms./Dr./Prof., Japanese: ■/■, Korean: ■, Thai: ■■■, Chinese: ■■, Vietnamese: Ông/Bà/Anh/Ch■). Family/given name splitting for both space-separated and CJK no-space names.
- **Calendar system detection:** Gregorian, JapaneseEra (■■/■■/■■/■■/■■), ThaiBuddhist (■.■.), ChineseLunar (■■), Hijri.
- **Currency normalisation:** Maps 9 currency symbols (\$, €, ¥, £, ■, ■, ■, ■, ■) to ISO 4217 codes. Handles European decimal format (1.000,00 € → 1000.00 EUR).
- **Address country detection:** JP postal (■), US ZIP+state, UK postcode (SW1A 1AA), DE 5-digit+Germany keyword.

4.7 Semantic Synonym Matching

The synonyms module provides an offline multilingual synonym map covering 40+ high-frequency business term groups across EN/JA/KO/ZH, including domain-specific vocabulary for legal, medical, and technical domains. The map is a compile-time constant — no allocations on the hot path. At query-expansion time, each query term is expanded with its synonyms before being sent to FTS5, increasing recall without requiring embeddings or a model.

- **Core coverage:** advertising, budget, decision, approval, deadline, meeting, contract, invoice, report, project, schedule, task, payment, vendor, database, deployment, security, compliance, training, hiring.

- **Legal domain:** lawsuit ↔ ■■■ ↔ ■■■, attorney ↔ ■■■ ↔ ■■■, verdict ↔ ■■■ ↔ ■■■, plaintiff ↔ ■■■, defendant ↔ ■■■, jurisdiction ↔ ■■■ ↔ ■■■, liability ↔ ■■■ ↔ ■■■, damages ↔ ■■■ ↔ ■■■.
- **Medical domain:** diagnosis ↔ ■■■ ↔ ■■■, prescription ↔ ■■■ ↔ ■■■, symptom ↔ ■■■, treatment ↔ ■■■ ↔ ■■■, patient ↔ ■■■, surgery ↔ ■■■ ↔ ■■■, allergy ↔ ■■■ ↔ ■■■, chronic ↔ ■■■.
- **Technical domain:** algorithm ↔ ■■■ ↔ ■■■, deployment ↔ ■■■ ↔ ■■■, framework ↔ ■■■ ↔ ■■■, repository ↔ ■■■ ↔ ■■■, debugging ↔ ■■■ ↔ ■■■, refactoring ↔ ■■■ ↔ ■■■, configuration ↔ ■■■ ↔ ■■■, authentication ↔ ■■■ ↔ ■■■.
- **Bidirectional:** if A is a synonym of B, B is a synonym of A.
- **Cross-lingual:** 'advertising' ↔ '■■■' ↔ '■■■' ↔ '■■■'.
- **FTS5 integration:** `expand_fts_query` builds OR-joined quoted term queries for safe FTS5 consumption.

4.8 SLM-Assisted Entity Refinement & Identifier Extraction

The `slm_refiner` module defines a trait-based architecture for refining entities typed as Unknown using context clues. On high-tier devices with an on-device SLM, the `SlmRefiner` dispatches to the `InferenceRouter` with the `RefineEntity` task for live SLM-based entity typing. On mid-tier devices, a `HeuristicRefiner` uses simple context clues. On low-tier devices, a `NoOpRefiner` returns Unknown for all candidates.

- **EntityRefiner trait:** `refine(text, candidates, config) → Vec`. Send + Sync for thread-safe usage.
- **SlmRefiner:** Live SLM dispatch via `InferenceRouter` with `InferenceTask::RefineEntity`. Parses SLM JSON output via `SlmEntityVerdict`. Falls back to Unknown on malformed output or dispatch failure.
- **NoOpRefiner:** Returns Unknown for all candidates. Used on low-tier devices and as fallback when SLM is unavailable.
- **HeuristicRefiner:** Context-based refinement using honorific prefixes (Mr./Ms./Dr./■■/■■/■■), organisation suffixes (Inc./Ltd./■■■■/■■■■/■■■■), organisation prefixes (■■■■/■■■■), currency symbols (\$/€/¥/£/■■/■■/■■/■■), and date-like patterns (digits with hyphens/slashes).
- **SLM-assisted identifier extraction:** `find_ambiguous_identifiers` detects tokens that look like identifiers but don't match standard regex patterns. `classify_ambiguous_identifiers_slm` dispatches to the SLM with `RefineEntity` task for classification. `SlmIdentifierVerdict` parses the SLM response.
- **RefinementConfig:** `max_entities_per_call` (default 32), `min_confidence` (default 0.7), `context_window_chars` (default 128).
- **Char-offset safe:** Converts character offsets to byte offsets for safe string slicing on multi-byte CJK text.

4.9 Cross-Cultural Calendar Conversion

The cultural module provides conversion functions that transform culture-specific dates into ISO 8601 proleptic Gregorian format:

- **Japanese era conversion:** `convert_japanese_era` parses ■■■/■■/■■/■■/■■ dates (e.g. '■■■6■■1■■15■■' → '2024-01-15'). Era start years are hardcoded for each era.

- **Thai Buddhist conversion:** `convert_thai_buddhist` subtracts 543 years from the Buddhist era (e.g. '■.■. 2567' → '2024'). Supports year-only and full date formats (YYYY-MM-DD, YYYY/MM/DD).
- **Chinese lunar conversion:** `convert_chinese_lunar` uses a pre-computed lookup table covering 2020–2030 (Lunar New Year, Lantern Festival, Mid-Autumn Festival for each year). For dates not in the table, interpolates from the closest known lunar new year using 29-day month approximation. Supports ■■/■■/■■ markers with Chinese month/day names (■■, ■■, ... ■■; ■■, ■■, ... ■■).
- **Hijri conversion:** `convert_hijri` uses the Kuwaiti algorithm to convert Hijri dates to Julian Day Number, then to Gregorian. Supports 'AH' and '■■' markers, year-only format, and month-name format (Ramadan, Shawwal, etc.) with 12 Hijri month names.
- **Dispatch function:** `convert_to_iso8601` auto-detects the calendar system and dispatches to the appropriate converter. Gregorian dates pass through unchanged.
- **ConvertedDate struct:** Carries the ISO date string, original calendar system, and original surface form for provenance tracking.

4.10 CJK Character Boundary Fix

Fixed a pre-existing bug in `evidence_store/src/importance.rs` where the `negation_window` (a character count) was used as a byte offset for string slicing. With CJK text (3-byte UTF-8 characters), this landed inside a multi-byte character, causing a panic. Added `floor_char_boundary` helper that walks backward to the nearest valid UTF-8 boundary before slicing.

5. Evidence Store & Encryption

SQLCipher with per-scope DEKs via HKDF. AEAD (XChaCha20-Poly1305) encryption. BLAKE3 content hashing enables cross-scope dedup. Noise-class goes to encrypted ring buffer (AEAD with random nonce per entry) — no evidence row, no FTS index. Priority-aware LRU eviction on configurable size cap: entries are evicted by lowest priority, then least recently accessed, then oldest id. `ring_buffer_touch` updates `last_accessed_at`, `ring_buffer_set_priority` updates priority, and `ring_buffer_read_window` touches all read entries.

Retrieval: FTS5 trigram search (language-agnostic), recency-ordered, hybrid FTS+embedding. Trigram tokenizer critical for CJK — no word segmentation needed.

5.1 Secure Deletion & SSD TRIM

The evidence store provides a complete physical erasure chain: logical delete → VACUUM → TRIM. `secure_vacuum` runs VACUUM with `secure_delete=ON` (zeroing freed pages) and restores the previous pragma setting. `secure_vacuum_after_forget` combines orphan purge + VACUUM, returning a `SecureDeletionReport`. `trim_database_file` issues a filesystem-level TRIM/discard for the database file:

- **macOS:** `F_PUNCHHOLE` fcntl punches holes in the file, allowing APFS to deallocate backing blocks.
- **Linux:** `FITRIM` ioctl trims all free blocks in the containing mount point (ext4/f2fs).
- **Other platforms:** No-op, returns `TrimReport { trimmed: false }`.
- **TrimReport:** Reports whether TRIM was issued and which method was used.

6. Synthesis Hierarchy

- **Channel** consumes raw evidence -> produces ChannelRecap.
- **Domain** consumes channel outputs only -> produces DomainSummary.
- **Tenant** consumes domain outputs + approved docs -> produces TenantSummary.

Enforced at type system level: ChannelOutput requires ChannelRecap; DomainSynthesisInput rejects raw ChannelMemoryObject. LlamaCppSynthesizer drives Bonsai-1.7B with GBNF grammar constraints. NoOpSynthesizer fallback for tests/low-tier.

7. Memory Decay & Retention

Retention score = 6 weighted inputs: pinning (0.5), retrieval freq (0.15), corroboration (0.10), contradiction (0.05), age (0.10), non-use (0.10). Pinned objects floor at 0.9.

7.1 Per-Culture DecayProfile

- **Default:** Critical 100y, Important 2y, Useful 30d, Noise 1d, Non-use 14d.
- **High-context** (ja,ko,zh,th,vi,id,ms): Important 3y, Useful 60d, Noise 3d, Non-use 21d.
- **Low-context** (en,de,fr,es): Important 1.5y, Useful 21d, Noise 12h, Non-use 10d.

7.2 Source-Deduplicated Corroboration Counting

The `MemoryObject` now tracks `corroboration_sources` — a list of source fingerprints that have already corroborated this object. The `record_corroboration_from_source` method deduplicates by source fingerprint: if the same author posts 3 times in the same Slack channel, the counter increments only once. A source fingerprint should be a stable identifier for the author of the corroborating evidence (e.g. 'slack:U12345', 'email:alice@example.com', 'github:octocat'), not the evidence row ID or channel ID.

- **Deduplication:** `record_corroboration_from_source` returns `true` if the source is new (counter incremented), `false` if already seen (only `last_accessed_at` refreshed).
- **Backward compatibility:** Legacy `record_corroboration` (no source fingerprint) still works — always increments, does not populate `corroboration_sources`.
- **serde(default):** `corroboration_sources` defaults to empty vec for deserialized legacy data.

8. Capabilities Implemented

Capability	Description	Key Files
Email Connector	HTML stripping, threading (Gmail threadId, Graph conversationId, attachments)	connectors/src/email.rs
CJK Lexicon	Japanese/Korean/Chinese business keyword enrichment	observation_engine/src/lexicon.rs
Live Bonsai Gating	Feature-gated + env-var-gated SLM integration	inference_router/tests/multilingual_bonsai.rs
Webhook Edge Cases	8 edge-case tests, empty validationToken fix	connectors/src/email.rs
SEA Platform Connectors	Shopee, Grab, Gojek regional connectors	connectors/src/shopee_regional.rs, grab.rs, gojek.rs
Per-Culture Decay	High-context vs low-context DecayProfile tuning	memory_manager/src/retention.rs
Negation Detection	28 negation cues, 4-token window before decision keyword	evidence_store/src/importance.rs
Result Clustering	BLAKE3 content_hash grouping, synthesis recap preference	evidence_store/src/retrieval.rs
Cross-Reference Graph	Threading metadata linking, bidirectional lookup	evidence_store/src/store.rs
Archived Object Purge	TTL-based (default 90d), pin protection, PurgeReport	memory_manager/src/purge.rs
Orphan Cleanup	Standalone detection + cleanup of body_store orphans	evidence_store/src/store.rs
Noise Promotion	Retroactive reclassification from ring buffer to evidence	evidence_store/src/store.rs
Per-Tier Synthesis Prompts	Divergent prompts/grammars/token budgets per tier	synthesis_engine/src/prompt_config.rs
Mixed-Language Detection	Script pre-pass, ScriptKind classification, CJK lane routing	evidence_store/src/script.rs
Secure Deletion + SSD TRIM	VACUUM + secure_delete + platform-specific TRIM ioctls	evidence_store/src/store.rs
Typed Entity Taxonomy	13 EntityType + 25 IdentifierKind variants	observation_engine/src/entity_types.rs
Industry-Specific Extractors	16 regex patterns: Finance, Healthcare, Legal, Mfg, Retail	observation_engine/src/entity_extractors.rs
Device-Tier-Aware Extraction	Low/Mid/High tiers mapped to DeviceTier	observation_engine/src/extractor.rs
Cross-Cultural Normalisation	Name order, honorifics, calendars, currency, addresses	observation_engine/src/cultural.rs
Semantic Synonym Matching	40+ multilingual synonym groups including legal/medical/technical	observation_engine/src/synonyms.rs
SLM-Assisted Entity Refinement	SlmRefiner (live SLM) + HeuristicRefiner + NoOpRefiner fallback	observation_engine/src/slm_refiner.rs
SLM-Assisted Identifier Extraction	Ambiguous identifier detection + SLM classification	observation_engine/src/entity_extractors.rs
SLM-Assisted Negation Detection	SemanticNegationDetector for complex constructions	evidence_store/src/importance.rs
Calendar Conversion	Japanese era + Thai Buddhist + Chinese lunar + Hijri → ISO 8601	observation_engine/src/cultural.rs
Priority-Aware LRU Ring Buffer	Priority + last_accessed_at eviction, touch + set_priority APIs	evidence_store/src/store.rs
Source-Deduplicated Corroboration	Per-source fingerprint tracking, dedup counting	memory_manager/src/object.rs
CJK Character Boundary Fix	floor_char_boundary for safe UTF-8 slicing	evidence_store/src/importance.rs

8.1 Capability Details

Negation Detection: The LexiconClassifier detects 28 negation cues (not, never, no, don't, without, cancel, reject, deny, refuse, decline, abandon, drop, kill, stop, undo, revert, rollback, void, withdraw, quash, overrule, overturn, repeal, rescind, revoke, nullify, invalidate, negate, countermand) within a 4-token window before decision keywords. When a negation cue is found, the classifier suppresses the decision/task classification, preventing 'decided NOT to' from being classified as a decision. 6 tests.

Result Clustering: HybridRetriever groups FTS+embedding results by BLAKE3 content_hash, deduplicating corroborating evidence. ClusteredRetrievalResult exposes a representative (highest-scoring member), cluster member IDs, content hash, and source count. Synthesis recaps are preferred as representatives. 8 tests.

Cross-Reference Graph: EvidenceStore supports add_cross_reference, get_cross_references, and get_reverse_cross_references backed by a cross_references table. Threading metadata (threadId, conversationId, message-id) is linked at ingestion time. 9 integration tests.

Archived Object Purge: memory_manager::purge provides PurgeConfig with configurable retention_days (default 90), pin protection, and PurgeReport. purge_archived transitions Archived objects past TTL to Deleted. 6 tests.

Orphan Cleanup: count_orphaned_bodies and purge_orphaned_bodies provide standalone detection and cleanup of body_store rows with zero remaining CEK wraps. 6 integration tests.

Noise Promotion: promote_from_ring_buffer re-ingests ring buffer content as evidence. reclassify stores importance override in a separate table, preserving append-only integrity. effective_importance resolves at read time. 8 tests.

Per-Tier Synthesis Prompts: SynthesisPromptConfig provides divergent system prompts, JSON grammar schemas, and token budgets per tier (Channel: 512, Domain: 1024, Tenant: 2048). 8 unit tests.

Mixed-Language Detection: detect_mixed_language returns MixedLanguageResult with all script families, dominant script, and needs_cjk_lanes flag. ScriptKind classifies into WhitespaceSegmented, CJKFamily, Symbol, Digit. 14 unit tests.

Secure Deletion + SSD TRIM: secure_vacuum runs VACUUM with secure_delete=ON. secure_vacuum_after_forget combines orphan purge + VACUUM. trim_database_file issues platform-specific TRIM (F_PUNCHHOLE on macOS, FITRIM on Linux). 8 integration tests.

Typed Entity Taxonomy: EntityType (13 variants) + IdentifierKind (25 variants across 5 industry domains). Observation struct gains entity_type and identifier_kind fields with backward-compatible serde defaults. 8 tests.

Industry-Specific Entity Extractors: 16 regex-based extractors covering Finance (IBAN w/ mod-97, ISIN, SWIFT/BIC, LEI, tickers), Healthcare (ICD-10, NDC), Legal (patents, case numbers), Manufacturing (SKU, serial numbers), Retail (ASIN, PO, invoice), and General (phone, IP, currency, measurements). OnceLock regex caching. 30 tests.

Device-Tier-Aware Extraction: EntityExtractionTier (Low/Mid/High) maps to DeviceTier. Low=lexicon-only, Mid=lexicon+patterns, High=lexicon+patterns+SLM refinement. 3 tier-gating tests.

Cross-Cultural Normalisation: Name order detection (FamilyFirst/GivenFirst), honorific stripping (7 languages), calendar system detection (5 systems), currency normalisation (9 symbols → ISO 4217), address country detection (4 countries). 23 tests.

Semantic Synonym Matching: Offline multilingual synonym map with 40+ business term groups across EN/JA/KO/ZH, including domain-specific vocabulary for legal (lawsuit, attorney, verdict, plaintiff, defendant, jurisdiction, liability, damages), medical (diagnosis, prescription, symptom, treatment, patient, surgery, allergy, chronic), and technical (algorithm, deployment, framework, repository, debugging, refactoring, configuration, authentication) domains. `expand_query` returns all synonyms for a term. `expand_fts_query` builds OR-joined FTS5 queries. `are_synonyms` checks bidirectional synonymy. 16 tests.

SLM-Assisted Entity Refinement: `EntityRefiner` trait with `SImRefiner` (live SLM dispatch via `InferenceRouter`), `HeuristicRefiner` (context-based), and `NoOpRefiner` (fallback). SLM-assisted ambiguous identifier extraction via `find_ambiguous_identifiers` and `classify_ambiguous_identifiers_slm`. `RefinementConfig` controls `max_entities_per_call`, `min_confidence`, `context_window_chars`. 18 tests.

SLM-Assisted Negation Detection: `SemanticNegationDetector` dispatches to `InferenceRouter` with `InferenceTask::DetectNegation` for SLM-based negation detection of complex constructions (e.g. 'we considered X but chose Y'). `NegationVerdict` parses SLM JSON output. `FallbackAdapter` provides heuristic negation detection when SLM is unavailable. 8 tests.

Calendar Conversion: `convert_japanese_era` (■■■■■■■■■■■■■■■■■■■■ → ISO 8601), `convert_thai_buddhist` (■■.■■. → ISO 8601), `convert_chinese_lunar` (■■■■■■■■■■ → ISO 8601, 2020–2030 lookup table + interpolation), `convert_hijri` (Kuwaiti algorithm → ISO 8601), `convert_to_iso8601` (auto-dispatch). `ConvertedDate` carries ISO date, original calendar, and surface form. 25 tests.

Priority-Aware LRU Ring Buffer: `ring_buffer_insert_with_priority` sets priority on insert. Eviction orders by priority ASC, `last_accessed_at` ASC, `id` ASC. `ring_buffer_touch` updates `last_accessed_at`. `ring_buffer_set_priority` updates priority. `ring_buffer_read_window` touches all read entries. Schema migration adds columns via ALTER TABLE. 5 tests.

Source-Deduplicated Corroboration: `record_corroboration_from_source` tracks source fingerprints and deduplicates. Same author posting 3x counts as 1. Legacy `record_corroboration` preserved. 4 tests.

9. Walkthrough: English Slack Decision

9.1 Content

```
@sarah "We decided to go with Vendor X for the payment integration. Deadline is March 15. @tom please start the API integration by end of week."
```

9.2 Ingestion

Slack connector emits DocumentCreated. detect_language identifies 'en'. Lexicon classifier matches 'decided' and 'deadline' -> Important class. Body encrypted inline with per-scope DEK. FTS5 trigram index populated in same transaction.

Done well: Language detection at write boundary stamps BCP-47 tag for downstream lexicons/FTS without re-detection.

Done well: Lexicon correctly identifies 'decided'/'deadline' as Important, keeping message out of Noise ring buffer.

Done well: SLM-assisted semantic negation detection (SemanticNegationDetector) handles complex constructions like 'we considered going with X but ultimately chose Y' by dispatching to the SLM with InferenceTask::DetectNegation, going beyond the heuristic keyword+window approach.

9.3 Roll-up

Observation engine extracts: Decision (Vendor X), Task (Tom: API integration), Entities (@sarah, @tom — typed as EntityType::Person, Vendor X — typed as EntityType::Organization), Date (March 15 — typed as EntityType::Date). On Mid/High tier devices, pattern-based extractors also scan for structured identifiers (IBAN, SWIFT/BIC, etc.). On High-tier devices, the HeuristicRefiner can refine any remaining Unknown entities using context clues. LlamaCppSynthesizer produces ChannelRecap with GBNF-constrained JSON SummaryBundle.

Done well: Typed entity taxonomy classifies @sarah and @tom as EntityType::Person, Vendor X as EntityType::Organization, and March 15 as EntityType::Date — replacing the previous flat Entity type.

Done well: SLM-assisted entity refinement via SlmRefiner dispatches to the InferenceRouter with InferenceTask::RefineEntity for live SLM-based entity typing on high-tier devices, with HeuristicRefiner fallback for mid-tier.

```
{
  "recap": "Team chose Vendor X for payment integration...",
  "decisions": ["Selected Vendor X as payment provider"],
  "active_tasks": ["Tom: start API integration"]
}
```

Done well: Synthesis hierarchy enforces channel recaps consume only raw evidence — domain cannot shortcut.

Done well: Supersession preferred over deletion — old versions retained for audit.

Done well: Per-tier synthesis prompts diverge: Channel gets concise recap instructions (512 tokens), Domain gets synthesis-focused prompts (1024 tokens), Tenant gets comprehensive roll-up prompts (2048 tokens) — each with its own JSON grammar schema.

9.4 Retrieval

Hybrid search: FTS5 trigram matches 'Vendor X' substrings. Embedding similarity adds semantic ranking. Results scope-filtered. retrieval_count incremented, feeding back into retention score.

Done well: Trigram tokenizer is language-agnostic — works for English, Japanese, Thai without word segmentation.

Done well: Result clustering groups corroborating evidence by BLAKE3 content_hash, deduplicating identical bodies and preferring synthesis recaps as cluster representatives.

Done well: Semantic synonym matching expands query terms with cross-lingual synonyms — querying 'budget' also matches '■■', '■■■', and '■■■■' without requiring embeddings.

9.5 Fading & Forgetting

Important class: 2y age half-life (default), 14d non-use half-life. Without retrieval, non-use decays to ~0 after ~70 days, but age component keeps score above archive threshold for 8+ years. Pinning floors at 0.9. forget_scope destroys DEK — all ciphertext unrecoverable.

Done well: Decay model well-calibrated: Important items survive long enough; Noise items evaporate from ring buffer quickly.

Done well: Cryptographic forgetting is thorough — DEK destruction makes all ciphertext permanently unrecoverable.

Done well: TTL-based purge transitions Archived objects to Deleted after a configurable retention period (default 90 days), with pin protection and PurgeReport tracking.

Done well: SSD TRIM automation completes the physical erasure chain: logical delete → VACUUM with secure_delete → filesystem TRIM (F_PUNCHHOLE on macOS, FITRIM on Linux).

10. Walkthrough: Japanese Email Threading

10.1 Content

```

#####: ##### - Q1#####
#####
Q1#####
#####: #####50#####
###: 215#####
###

```

10.2 Ingestion

Graph webhook fires. Language detection: 'ja' from CJK trigrams. Japanese lexicon: ■■ matches critical keywords -> Critical class (no passive decay), fetch_content surfaces conversationId for threading.

Done well: Enriched CJK lexicon correctly identifies ■■ (approval) as Critical — formal Japanese business communications get highest retention.

Done well: Threading metadata (conversationId) enables conversation correlation across replies.

Done well: Mixed-language detection provides `detect_mixed_language`, a script-detection pre-pass that identifies all Unicode script families in text and flags CJK-family content for trigram+bigram FTS lane routing.

10.3 Roll-up

Japanese lexicon extracts: Decision (50■■■ budget allocation, matched by ■■■■■), Task (■■■ request), Entities (■■■■■, ■■■ — typed as EntityType::Person), Date (2■■15■ — typed as EntityType::Date), Currency (50■■■ — typed as EntityType::Currency). If the date were written in Japanese era format (e.g. ■■■6■■2■■15■), `convert_to_iso8601` would convert it to '2024-02-15'. ChannelRecap synthesised with SLM.

Done well: Typed entity taxonomy stamps each Entity observation with a specific EntityType (Person, Date, Currency) instead of a generic Entity — downstream consumers can apply type-specific processing without re-parsing.

Done well: Cross-cultural normalisation detects Japanese name order (FamilyFirst for 田中太郎), strips honorifics, and converts 2024.01.01 calendar dates to ISO 8601 — enabling culturally-aware entity processing.

Done well: Currency extraction recognises ¥/¥ amounts and normalises to ISO 4217 (JPY), handling both Western and Asian currency formats.

10.4 Retrieval

Query '■■■■■■■■■■' triggers trigram search: '■■■', '■■■', '■■■' etc. Matches indexed email body without word boundaries — crucial for Japanese (no spaces). Query '■■■' (advertising expense) also matches via the offline synonym map, which maps ■■■ ↔ ■■ ↔ advertising.

Done well: Trigram tokenizer sidesteps need for MeCab/Kuromoji morphological analysers — no binary weight, no per-language config.

Done well: Semantic synonym matching bridges cross-lingual vocabulary gaps without embeddings — '■■■■' for ■■■■■■ now matches via the synonym map.

10.5 Fading

Critical class: 100-year half-life — effectively no passive decay. Decay sweep exempts Critical from archival. Only explicit deprecation/supersession/forgetting removes it. Superseded objects archive after 90d TTL.

Done well: Critical-class immunity from passive decay is correct — regulatory approvals shouldn't disappear from non-use.

Done well: High-context DecayProfile (for_language('ja')) extends Important-class half-life to 3y, reflecting higher contextual weight in Japanese business culture.

11. Walkthrough: Noise-class Chat Reaction

11.1 Content

+1 ■

11.2 Ingestion

Language detection returns None (too short). Lexicon: '+1' and '■' in noise_tokens -> Noise class. Storage router selects RingBuffer: AEAD-encrypted, no evidence row, no FTS index, no embedding. Priority-aware LRU eviction on configurable size cap — entries are evicted by lowest priority, then least recently accessed.

Done well: Noise never enters evidence plane — saves storage, encryption overhead, FTS bloat. Ring buffer provides short-term context.

Done well: Ring buffer entries are AEAD-encrypted with random nonce per entry — same privacy guarantee as other classes.

Done well: Noise token list is configurable per-tenant.

11.3 No Roll-up, No Retrieval, No Fading

Noise doesn't participate in synthesis, isn't searchable, doesn't enter decay state machine. Lifecycle managed entirely by ring buffer eviction. Correct behaviour — knowledge hierarchy shouldn't be polluted by social reactions.

Done well: Noise path is extremely cheap: no FTS, no embedding, no synthesis. Keeps ingest fast in high-volume channels.

Done well: Priority-aware LRU eviction keeps recently accessed and high-priority noise entries longer than pure FIFO, improving short-term context utility.

Done well: Noise promotion allows retroactive reclassification: `promote_from_ring_buffer` re-ingests ring buffer content as a proper evidence row, and reclassify stores an importance override in a separate table (preserving append-only evidence integrity).

12. Walkthrough: Cross-source Corroboration

12.1 Content

- **Slack (user U123):** 'We decided to use PostgreSQL for the database.'
- **Slack (user U456):** 'Re: Database selection — PostgreSQL confirmed'
- **GitHub (user octocat):** 'Database decision: PostgreSQL. See slack thread for context.'

12.2 Ingestion

Each source triggers its own connector. All classified Important ('decided'/'confirmed'/'decision'). All get encrypted evidence rows with FTS5 indexes.

12.3 Roll-up

Observation engine extracts decision from each. ChannelRecap includes the decision. MemoryObject created with `corroboration_count=3` (three unique source fingerprints: `slack:U123`, `slack:U456`, `github:octocat`). Retention score's corroboration component saturates at $3/3=1.0$, contributing full 0.10 weight.

Done well: Cross-source corroboration is powerful — a decision backed by Slack+GitHub is more trustworthy. Retention model correctly rewards this.

Done well: Source-deduplicated corroboration counting ensures that if user U123 posts 3 follow-up messages confirming the same decision, the counter still reflects 2 unique sources (U123 + U456 + octocat), not 5.

Done well: Cross-reference graph links evidence rows via threading metadata. `add_cross_reference` stores key-value pairs and `get_cross_references` / `get_reverse_cross_references` enable bidirectional traversal.

12.4 Retrieval

Query 'PostgreSQL' returns all three evidence rows via FTS5. Hybrid path may also surface ChannelRecap. User sees multiple perspectives — increases confidence.

Done well: Result clustering groups corroborating results by `content_hash`, presenting a single representative (highest-scoring or synthesis recap) with cluster member IDs and source count.

12.5 Fading

Important class, 2y half-life. Corroboration=3 contributes 0.10 indefinitely (non-decaying). If superseded ('switching to MongoDB'), old synthesis transitions to Superseded -> Archived after 90d.

Done well: Corroboration is non-decaying — multi-source confirmation shouldn't fade with time. Correct design.

13. Walkthrough: Cryptographic Forgetting

13.1 Scenario

A user leaves the company. Their personal scope (1:1 messages, private emails, observations) must be permanently destroyed. Host calls `forget_scope` with the scope UUID.

13.2 What Happens

- **Step 1:** Scope DEK cryptographically destroyed. Tombstone inserted.
- **Step 2:** All evidence rows deleted. Encrypted bodies become unrecoverable ciphertext.
- **Step 3:** MemoryObjects for scope deleted.
- **Step 4:** Synthesis objects for scope deleted.
- **Step 5:** FTS5 index entries removed.
- **Step 6:** Connector instances dropped from in-memory map.
- **Step 7:** Scope tombstoned — future operations return `NotFound`.
- **Step 8:** `secure_vacuum_after_forget` purges orphans + `VACUUM` with `secure_delete=ON`.
- **Step 9:** `trim_database_file` issues filesystem TRIM (`F_PUNCHHOLE` on macOS, `FITRIM` on Linux).

After forgetting, encrypted bytes may remain on disk but without DEK they are unrecoverable. `VACUUM/checkpoint` overwrites physical pages with zeros. TRIM tells the SSD controller to erase the freed flash blocks, preventing forensic recovery of stale ciphertext.

Done well: Multi-layered: key destruction + row deletion + tombstone. Forensic disk image won't help without the destroyed DEK.

Done well: Tombstone prevents data resurrection — webhooks/syncs after forgetting silently drop data.

Done well: Shared ciphertext orphan cleanup: `count_orphaned_bodies` and `purge_orphaned_bodies` provide standalone detection and cleanup of `body_store` rows with zero remaining CEK wraps.

Done well: Secure deletion automation: `secure_vacuum` runs `VACUUM` with `secure_delete=ON` (zeroing freed pages) and restores the previous pragma setting.

Done well: SSD TRIM automation: `trim_database_file` issues platform-specific TRIM ioctls to physically erase freed flash blocks, completing the erasure chain.

14. Overall Critique & Remaining Areas

14.1 Strengths

- **Privacy-first architecture:** Per-scope DEKs, AEAD encryption, cryptographic forgetting. Data is encrypted at rest with keys scoped to individual contexts. Ring buffer entries are also AEAD-encrypted.
- **Language-agnostic retrieval:** FTS5 trigram tokenizer works across CJK and European languages without morphological analysers or per-language config.
- **Lexicon-first extraction:** Works offline, no model required. Enriched CJK lexicons cover formal/business register.
- **Typed entity taxonomy:** 13 EntityType variants + 25 IdentifierKind variants provide structured sub-classification for every Entity observation, enabling downstream industry-specific processing without re-parsing content strings.
- **Industry-specific identifier extraction:** 16 regex-based extractors covering Finance, Healthcare, Legal, Manufacturing, Retail, and General domains. IBAN mod-97 checksum validation and IPv4 octet validation ensure high precision.
- **Device-tier-aware extraction:** EntityExtractionTier (Low/Mid/High) maps to DeviceTier, ensuring low-end devices skip pattern matching while mid/high devices get full identifier extraction.
- **Cross-cultural normalisation:** Name order detection, honorific stripping (7 languages), calendar system detection and conversion (Japanese era, Thai Buddhist, Chinese lunar, Hijri → ISO 8601), currency normalisation (9 symbols → ISO 4217), and address country detection enable culturally-aware entity processing.
- **Semantic synonym matching:** Offline multilingual synonym map with 40+ business term groups across EN/JA/KO/ZH, including domain-specific legal, medical, and technical vocabulary, provides cross-lingual query expansion without embeddings — critical for low-tier devices.
- **SLM-assisted entity refinement:** Trait-based architecture with SlmRefiner (live SLM dispatch), HeuristicRefiner (context clues), and NoOpRefiner (fallback) enables graceful degradation across device tiers. SLM-assisted ambiguous identifier extraction and semantic negation detection extend SLM coverage. Char-offset safe for CJK text.
- **Source-deduplicated corroboration:** Per-source fingerprint tracking ensures corroboration count reflects unique authors, not message volume. Backward-compatible with legacy counting.
- **Hierarchy-enforced synthesis:** Type-system guarantees that domain synthesis cannot consume raw evidence, tenant cannot consume channel objects. Clean separation of concerns.
- **Retention model:** Six weighted inputs with pinning floor, per-culture DecayProfile tuning, non-decaying source-deduplicated corroboration. Well-calibrated half-lives per importance class.
- **Connector framework:** 120+ connectors with unified trait, watermark cursor atomicity, webhook support, mockable transport for testing.
- **Supersession over deletion:** Old synthesis versions retained for audit. CRDT-compatible design.
- **Priority-aware ring buffer:** LRU eviction with priority support replaces FIFO. Entries are evicted by lowest priority, then least recently accessed, then oldest id. ring_buffer_touch and ring_buffer_set_priority APIs enable dynamic priority management.

- **Complete erasure chain:** Logical delete → VACUUM with secure_delete → SSD TRIM (F_PUNCHHOLE/FITRIM). Three layers of defense against forensic recovery.

14.2 Remaining Areas for Future Enhancement

- Live SLM integration testing: The SImRefiner, SemanticNegationDetector, and SLM-assisted identifier extraction are architected for live SLM dispatch but hermetic tests use FallbackAdapter. End-to-end testing with a live Bonsai model would validate accuracy on real-world text.
- Chinese lunar calendar coverage: The lookup table covers 2020–2030. Extending beyond this range or implementing algorithmic conversion (astronomical leap month rules) would support a wider date range.
- Hijri calendar accuracy: The Kuwaiti algorithm is an approximation. For applications requiring exact Hijri dates (e.g. religious observances), a more precise algorithm or Umm al-Qura lookup table may be needed.

End of Report